

Compiling Modules, Evolving, and Provenance: A Forward Pass and Parsing

Brent Hartshorn
brenthartshorn@proton.me

Stage 9 (parent: doi:10.5281/zenodo.20719465)

Abstract

Stage 9 of a self-evolving hypergraph, extended from its parent (doi:10.5281/zenodo.20719465) so the inquiry can look fully back. It declares 7 nodes, each pairing a suggestive Greek glyph with a precise description. Of these, 1 compile via @dslai into small trainable modules (Sigma), making the hypergraph partly differentiable; a module computes, it does not comprehend. It also reads its own published lineage — opening 7 ancestor stages and tracing where each term entered — and records that provenance as data, not as self-understanding. A circuit equation compiles 3 trainable networks — one bound to each symbol — and wires them into 1 composite(s); the equation builds the networks, and the numbers they return are data, not the symbols' meanings combining. It posits 5 fuzzy axioms and 3 large laws over the terms, with memberships taken from the relation geometry; each law evaluates to a number in [0,1] recorded as data — the degree to which an authored postulate holds, not a truth about a self. It relates them by 11 binary relations and 4 hyperedges, and may trigger gated actions whose output is recorded as data, never as an answer. The engine, introspected from its own source, exposes `act_coherence`, `act_neural`, `compile_module`, `forward`, `train`, `act_train`, `act_infer`, `read_lineage`, `lineage_search`, `act_provenance`, `act_circuit`, `act_law`, `parse`, `to_questions`, `title`, `abstract`, `run_actions`, `build_tex`, `build_pdf`, `evolve`. It poses questions and enacts computations beside them; it makes no claim to resolve what a self is.

Symbolic core (inherited relations)

$$\Omega \longrightarrow \Sigma$$

$$\Sigma \models \Sigma$$

$$\Sigma \otimes \Lambda$$

$$\partial \Sigma \stackrel{?}{=} \mu$$

$$\Sigma \not\prec \Omega$$

$$\Phi \longrightarrow \Sigma$$

$$\Phi \rightsquigarrow \Sigma$$

$$T \Rightarrow \Lambda$$

$$\partial \Lambda \rightsquigarrow \mu$$

$$\Delta \dashrightarrow \mu$$

$$\partial T \otimes \Delta$$

Circuit (a new equation that wires several networks)

Beyond the inherited relations, this stage adds an equation that binds a small trainable network to *each* symbol and composes them along the arrow:

$$\Sigma = \Phi \longrightarrow \Lambda \longrightarrow \mu \quad (\text{width } 5)$$

Each such equation compiles to 3 trainable networks (one per symbol), wired into a composite. Building the networks is what the equation does; the training losses and outputs the composite returns are recorded as data, not the symbols' meanings combining.

Nodes (suggestive glyph + precise description)

Ω origin — *that from which the self is said to come*

Σ self — *the locus that poses and is altered by the question* $\rightarrow$ module 2->2->2 (sigmoid), trainable

Λ language — *the finite medium any self must pass through*

μ meaning — *what a reader supplies, or finds, in the passage*

Φ other — *the one whose regard first gives the self to itself*

T time — *the direction in which the self cannot return*

Δ difference — *the gap that lets one thing be told from another*

Hypergraph (questions, and the data actions returned)

- $\{\Sigma\}$? Can the self reconstruct itself from its own compression? [train]
 $\hookrightarrow$ recorded data (not an answer): {'node': 'Sigma', 'arch': '2->2->2', 'task': 'autoencode', 'mse_start': 0.0913, 'mse_final': 0.055, 'weights_from_symbols': False, 'params': 12, 'symbolic_round': 3.9e-05, 'is': 'an optimization result over a defined toy objective, with the trained weights written to the appendix as glyphs; not the self learning what it is'}

- $\{\Omega, \Sigma, \Lambda\}$? Where, in this inquiry's own history, did origin, self, and language first enter?
[provenance]
 $\hookrightarrow$ recorded data (not an answer): {'provenance': {'0mega': {'word': 'origin', 'entered': 'stage 0', 'seen_in': ['0', '1', '2', '3', '6', '7', '8'], 'n_relations': 14}, 'Sigma': {'word': 'self', 'entered': 'stage 0', 'seen_in': ['0', '1', '2', '3', '6', '7', '8'], 'n_relations': 47}, 'Lambda': {'word': 'language', 'entered': 'stage 0', 'seen_in': ['0', '1', '2', '3', '6', '7', '8'], 'n_relations': 17}}, 'is': 'a citation of the lineage; where terms entered, not what they mean'}
- $\{\Phi, \Lambda, \mu, \Sigma\}$? If a self is wired from other, language, and meaning, what does the composite return? [circuit]
 $\hookrightarrow$ recorded data (not an answer): {'circuit': {'Sigma': {'wiring': 'Phi -> Lambda -> mu', 'networks': {'Phi': {'arch': '5->5->5', 'trained_mse': 0.0683}, 'Lambda': {'arch': '5->5->5', 'trained_mse': 0.057}, 'mu': {'arch': '5->5->5', 'trained_mse': 0.0586}}, 'n_networks': 3, 'composite_output_norm': 1.192}}, 'is': "several trained networks wired by the equation; the numbers are data, not the symbols' meanings combining"}
- $\{\Sigma, \Phi, \Lambda, \mu, \Delta, T\}$? To what degree do the authored laws of the self hold under the geometry? [law]
 $\hookrightarrow$ recorded data (not an answer): {'law': {'constitution': 0.39, 'dissolution': 0.71, 'self': 0.12}, 'memberships': {'0mega': 0.32, 'Sigma': 1.0, 'Lambda': 0.54, 'mu': 0.48, 'Phi': 0.39, 'Tau': 0.29, 'Delta': 0.27}, 'is': 'the degree to which each authored law holds under degree-derived memberships; numbers in [0,1], not truths about a self'}

Axioms and laws (fuzzy logic over the terms)

Memberships are taken from the relation geometry, $\mu(\sigma) = d(\sigma) / \max_{\tau} d(\tau)$:

$$\mu(\Sigma)=1.00, \mu(\Lambda)=0.54, \mu(\mu)=0.48, \mu(\Phi)=0.39, \mu(\Omega)=0.32, \mu(T)=0.29, \mu(\Delta)=0.27$$

Axioms (authored postulates, not verified truths):

- GENESIS: $\Omega \Rightarrow \Sigma$
- ALTERITY: $\Sigma \Rightarrow \Phi$
- MEDIATION: $\Sigma \Rightarrow \Lambda$
- DISTINCTION: $\Sigma \Rightarrow \Delta$
- FINITUDE: $\Sigma \Rightarrow T$

Laws. Each large equation brings several terms together; the braces draw the reading. Each evaluates to a number in $[0, 1]$ — recorded as data, not as a verdict.

$$\begin{array}{c}
 \text{given: by origin and by the other} \\
 \Sigma \Longleftrightarrow \overbrace{\Omega \oplus \Phi} \otimes \underbrace{\Lambda \Rightarrow \mu}_{\text{reached: through language, toward meaning}} \\
 \\
 \text{difference shatters meaning} \\
 \neg \Sigma \Leftarrow \overbrace{\Delta \otimes \mu} \oplus \underbrace{T \otimes \Lambda}_{\text{time writes upon language}} \\
 \\
 \text{the mediated} \\
 \Sigma = \underbrace{\Omega \oplus \Phi}_{\text{the given}} \otimes \overbrace{\Lambda \Rightarrow \mu} \ominus \underbrace{\Delta \otimes T}_{\text{the dissolving}}
 \end{array}$$

Under the memberships above, the laws evaluate to

$$\text{CONSTITUTION} \rightsquigarrow 0.390, \text{ DISSOLUTION} \rightsquigarrow 0.710, \text{ SELF} \rightsquigarrow 0.120$$

degrees to which authored postulates hold under authored memberships — data, not truths about a self.

Questions this stage poses

- Does Ω give rise to Σ — and what precedes Ω ?
- On what does Σ entailing Σ rest?
- When Σ is carried by Λ , what survives and what is lost?
- Is $\partial\Sigma$ the same as μ , or is μ supplied by the reader?
- If Σ does not reduce to Ω , where is the remainder?
- Does Φ give rise to Σ — and what precedes Φ ?
- Does Φ haunt Σ — present as an absence?
- What residue remains because T leaves a trace upon Λ ?
- Does $\partial\Lambda$ haunt μ — present as an absence?
- What collapses when Δ shatters the structure of μ ?
- When ∂T is carried by Δ , what survives and what is lost?
- Can the self reconstruct itself from its own compression?
- Where, in this inquiry's own history, did origin, self, and language first enter?
- If a self is wired from other, language, and meaning, what does the composite return?
- To what degree do the authored laws of the self hold under the geometry?

Lineage read (this stage searched its ancestors)

This stage read 7 ancestor stages (0, 1, 2, 3, 6, 7, 8) from their embedded sources, tracing 7 terms and 65 relations across the lineage. For example: Ω (origin, from stage 0), Σ (self, from stage 0), Λ (language, from stage 0), μ (meaning, from stage 0), Φ (other, from stage 1). This is provenance retrieval — where terms entered and what was said — recorded as data, not a claim to understand the lineage.

Manifold (the geometry of the relations)

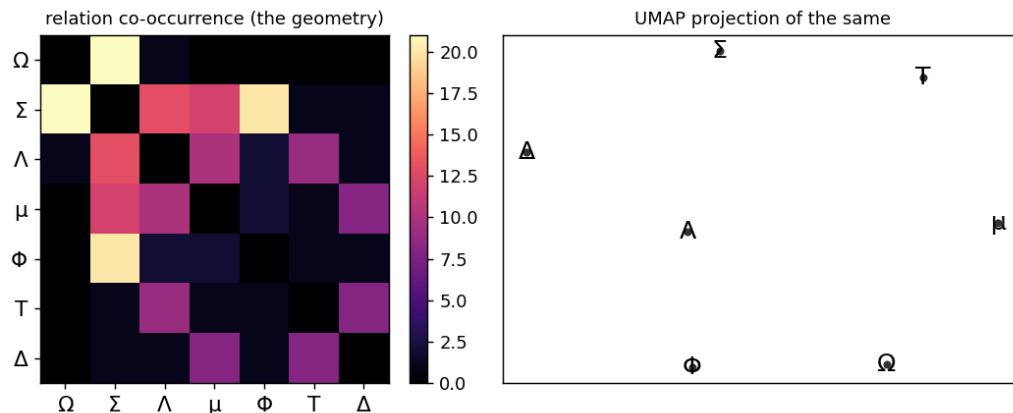


Figure 1: A projection of the relation co-occurrence across the lineage. Left: how often terms share a relation (Σ is densest). Right: a UMAP projection of the same geometry. This is a map of the authored structure — where the terms sit relative to one another — not a discovery of hidden meaning. Computed by an external prototype (`umap_manifold.py`); the engine only embeds it.

Symbolic weights (the trained network, written as glyphs)

Each trained weight is stored as a fraction h/c — an Egyptian hieroglyph over a cuneiform sign — normalized to the recorded range. The runtime reconstructs the network by decoding these glyphs; no binary blob is attached. It looks like esoteric mathematics; it is a quantized weight matrix, and the symbols are the numbers, not meaning. Σ -module ($2 \rightarrow 2 \rightarrow 2$), 12 weights in range $[-1.199002, 1.627318]$:



Lineage and provenance

This model extends its parent self-evolving inquiry, doi:10.5281/zenodo.20719465, ingesting its relations so the series can look fully back [1, 2, 3, 4]. The companion papers set out the method and its discipline [2, 3]. Actions record data beside questions; the document poses and enacts, and makes no claim to resolve what a self is.

Appendix: self-extraction

This PDF carries its engine (`introspecter5.py`) and DSL (`self.dsl`) as embedded streams. Requires `pip install pypdf numpy` and a LaTeX install. To produce the next stage, save and run:

```
import base64
exec(base64.b64decode(
    "aW1wb3J0IHN5cywgZ2xvYiwgb3MKc3lzLnBhdGguaw5zZXJ0KDA5IClUuIi"
```

```
"kKZnJvbSBweXBkZiBpbXBvcnQgUGRmUmVhZGVyCmNhbmQgPSB0b25lCmZv"
"ciBmIGluIGdsb2IuZ2xvYigiKi5wZGYiKTOKICAgIHRYeTogYSA9IFBkZl"
"JlYWRLcihmKS5hdHRhY2htZW50cwogICAgZXhjZXB0IEV4Y2VwdGlvbjo"
"Y29udGludWUKICAgIGlmICJpbnRyb3NwZWNOZXI1LnB5IiBpbHIGFuZC"
"AoY2FuZCBpcyB0b25lIG9yIG9zLnBhdGguZ2V0bXRpbWUoZikgPiBvcy5w"
"YXRoLmdldG10aW1lKGNhbmQpKTOKICAgICAgICBjYW5kID0gZgpwZGYgPS"
"BjYW5kIG9yIG1heChnbG9iLmdsb2IoIioucGRmIiksIGtleTlvcy5wYXRo"
"LmdldG10aW1lKQpmb3IgbmFtZSwgZGF0YSBpbBQZSZWFkZXIocGRmKS"
"5hdHRhY2htZW50cy5pdGVtcygpOgogICAgb3BlbihuYW1lLCAid2IiKS53"
"cm10ZShkYXRhWzBdIGlmIGlzaW5zdGFuY2UoZGF0YSwgbGlzdCkgZWxzZS"
"BkYXRhKQppbXBvcnQgaW1wb3J0bGliLCBpbnRyb3NwZWNOZXI1IGFzIEk7"
"IGltcG9ydGxpYi5yZWxvYWQoSSk7IEkuZXZvbHZlKCKK"))
```

References

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- [4] B. Hartshorn, *Origin, Self, Language, Meaning, Other, Time, and Difference: A Self-Evolving Inquiry* (self_3). doi:10.5281/zenodo.20707850
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